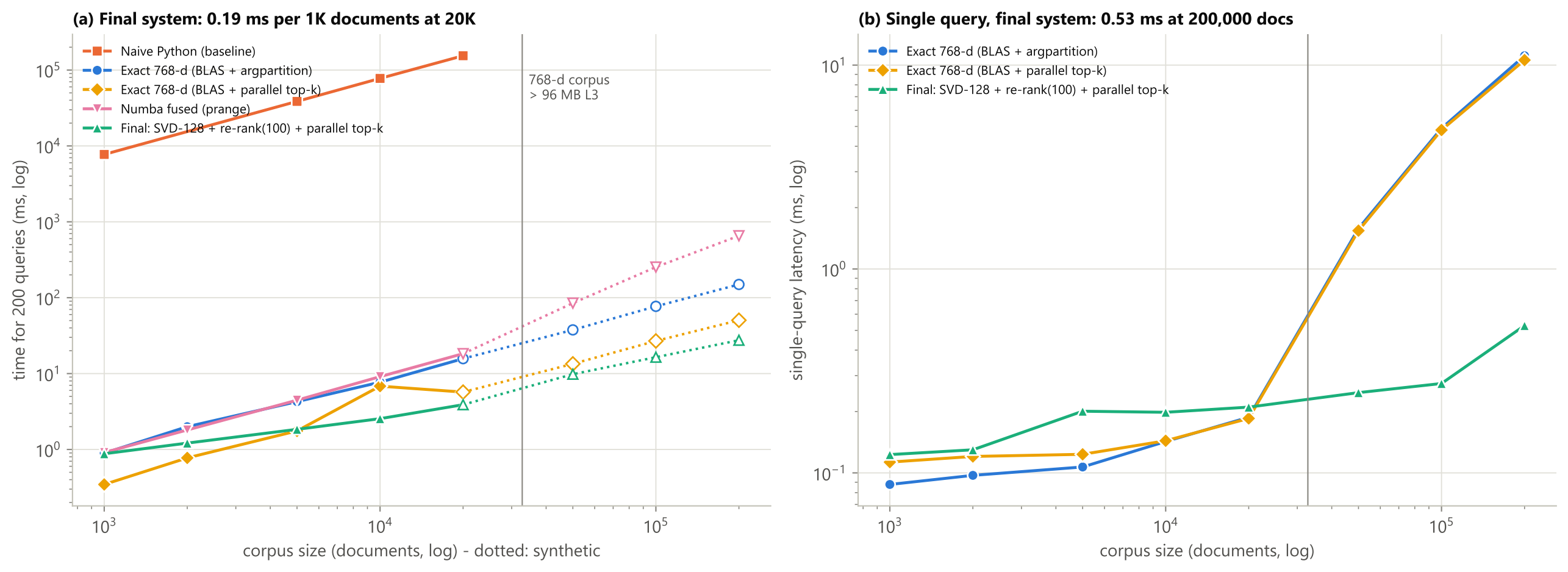
(a) Final system: 0.19 ms per 1K documents at 20K
Naive Python (baseline)
Exact 768-d (BLAS + argpartition)
Exact 768-d (BLAS + parallel top-k)
Numba fused (prange)
Final: SVD-128 + re-rank(100) + parallel top-k
768-d corpus
> 96 MB L3
time for 200 queries (ms, log)
corpus size (documents, log) - dotted: synthetic
(b) Single query, final system: 0.53 ms at 200,000 docs
Exact 768-d (BLAS + argpartition)
Exact 768-d (BLAS + parallel top-k)
Final: SVD-128 + re-rank(100) + parallel top-k
single-query latency (ms, log)
corpus size (documents, log)